

MATH3171 (Mathematical Biology III)
YEAR 2025–26

MICHAELMAS REVISION CLASS

This sheet contains what was written on the board in the revision class on Monday 27 April 2026.

1 General exam advice

(a) Format:

- Section A: Q1–4. 10 marks each.
- Section B: Q5–8. 15 marks each.
- No choice.

(b) Within each sections, questions do not increase in difficulty.

(c) Think about order (read it all first).

(d) Think about timing – how long to give to each question.

(e) Read the whole question. Maybe something at the end will tell you a hint or an assumption you can use.

(f) Lost? Can you do part (b) even if you can't do part (a)? This might require having to write down an assumption but that's OK.

(g) Move on! Use the mark distribution for time guidance.

(h) Use lots of paper! (Ask for more!)

(i) no rough paper

(ii) I will mark everything unless you cross it out

(iii) new question, new page

(iv) I want to give you marks, so make it easy for me! Write neatly!

2 Math Bio exam advice

(a) Do last year's paper and the practice paper already! (Don't save it.)

(b) Use bullet points.

(c) Draw pictures.

3 Rough course outline

- (a) 1D population models (time-dependent only)
 - (i) equilibria (feasible/missible)
 - (ii) stability (drawing the phase space graph of dx/dt against x is nearly always the best way to determine stability in 1D)
 - (iii) interpretation (drawing a picture is again the best method)
- (b) 1D & 2D population models
 - (i) nondimensionalisation
 - (ii) linear stability analysis formally (with ε)
 - (iii) bifurcations (changes in the number or behaviour of equilibria): drawing bifurcation diagrams for 1D models and interpreting them
- (c) 2D population models
 - (i) linear stability analysis shortcuts (just writing down $\mathbf{J}$, det/tr rules)
 - (ii) phase portraits (draw and interpret)
- (d) 3D population models
 - (i) e.g. SIR
 - (ii) reduction to 2D
- (e) (advection–)diffusion
 - (i) derivations
 - (ii) separation of variables (bounded domain)
 - (iii) Fourier transforms and Green's functions (unbounded domain)
- (f) travelling waves
 - (i) existence (between which equilibria?)
 - (ii) stability (indicating permissible wavespeeds)
- (g) PDE stability (the start)

4 How an exam question is written

- The little bolt-on (e.g. interpret, non-dimensionalise, consider a simpler version)
- The meat
- Something that tests understanding

5 Bifurcations

You need to be able to go between

single-species model $\longleftrightarrow$ bifurcation diagram.

For example,

$$\frac{dx}{dt} = x - k$$

has equilibrium $x_0 = k$. Is it stable or unstable?

Drawing the phase portrait shows that k is unstable.

So drawing the bifurcation diagram of x_0 against k , you get a dotted line of slope 1 which passes through the origin.

You should practice this for single-species (aka 1D) models that we've seen.

If you saw a bifurcation diagram, can you write down what dx/dt should be?

6 Nondimensionalisation

e.g. **Exam 2022 Q2**:

Convert

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + ru \left(1 - \frac{u}{K}\right), \quad x \in [0, L]$$

into, after hats have been removed,

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \alpha u(1 - u), \quad x \in [0, 1].$$

Normal procedure:

$$u = U\hat{u}, \quad x = X\hat{x}, \quad t = T\hat{t},$$

which includes substituting in the interval definition:

$$\begin{aligned} \frac{U}{T} \frac{\partial^2 \hat{u}}{\partial \hat{t}^2} &= \frac{DU}{X^2} \frac{\partial^2 \hat{u}}{\partial \hat{x}^2} + rU\hat{u} \left(1 - \frac{U\hat{u}}{K}\right), \quad X\hat{x} \in [0, L] \\ \implies \frac{\partial^2 \hat{u}}{\partial \hat{t}^2} &= \frac{DT}{X^2} \frac{\partial^2 \hat{u}}{\partial \hat{x}^2} + rT\hat{u} \left(1 - \frac{U\hat{u}}{K}\right), \quad \hat{x} \in \left[0, \frac{L}{X}\right] \end{aligned}$$

Starting with the interval definition, we get $X = L$. The diffusion term gives us $T = L^2/D$. Therefore $\alpha = rT = rL^2/D$.

If we have a boundary condition, e.g.

$$u(L, t) = \beta,$$

Think about what this means. It means

$$\begin{aligned} u &= \beta \text{ when } x = L \text{ and } \forall t \\ \implies U\hat{u} &= \beta \text{ when } X\hat{x} = L \text{ and } \forall T\hat{t} \\ \implies \hat{u} &= \frac{\beta}{U} \text{ when } \hat{x} = \frac{L}{X} \text{ and } \forall \hat{t}, \end{aligned}$$

i.e.

$$\hat{u}\left(\frac{L}{X}, \hat{t}\right) = \frac{\beta}{U}.$$

7 Phase portraits

e.g. **Exam 2022 Q8.5:**

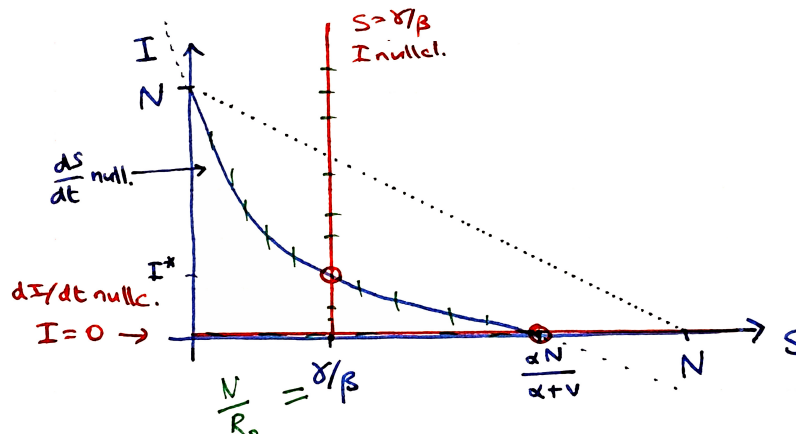
System is

$$\begin{aligned}\frac{dS}{dt} &= \alpha(N - S - I) - \beta IS - \nu S, \\ \frac{dI}{dt} &= \beta IS - \gamma I.\end{aligned}$$

It has nullclines

$$\begin{aligned}0 &= \alpha(N - S - I) - \beta IS - \nu S & \implies I &= \frac{\alpha N - (\alpha + \nu)S}{\alpha + \beta S}, \\ 0 &= I(\beta S - \gamma) & \implies I &= 0, S = \frac{\gamma}{\beta}.\end{aligned}$$

Remember that nullclines can be curves, not just straight lines.



To draw the I nullcline here, see that for small S , the equation for the nullcline looks like $-S$, and for large S , it looks like a constant.

There is no requirement in the question to put in trajectories, but if you wanted to, you could

- (i) put in numbers to check a region, or
- (ii) try $(\varepsilon, \varepsilon)$,

e.g.

$$\frac{dS}{dt} \sim \alpha N > 0, \quad \frac{dI}{dt} \sim -\gamma \varepsilon < 0,$$

then use the fact that signs flip over the relevant nullclines.

8 Travelling waves

Travelling waves go from equilibrium to equilibrium. Which way? Think domino-o-oes.

For $x - ct$ you have a rightward travelling wave:

stable $\rightarrow$ unstable

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For $x + ct$ you have a leftward travelling wave:

unstable $\leftarrow$ stable

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